

Fit risoluzione in p_T vs $|\eta|^{\text{truth}}$: r_0 libero vs $r_0 = 0$

$$\sigma_{p_T}/p_T = \sqrt{r_0^2/p_T^2 + r_1^2 + (r_2 \times p_T)^2} \text{ -- tutti i sample combinati (Z + Z')}$$

$ \eta $ bin	n pt	r_0 free	χ^2/ndf free	χ^2/ndf fix $r_0=0$	r_2 [GeV^{-1}]
$0.0 \leq \eta < 0.1$	12	0.000 ± 0.030	1545.75	1391.18	$0.186 \pm 0.001 \ (\times 10^{-3})$
$0.1 \leq \eta < 1.05$	12	0.000 ± 0.004	17036.76	15333.09	$0.112 \pm 0.000 \ (\times 10^{-3})$
$1.05 \leq \eta < 1.3$	12	0.000 ± 0.069	633.20	569.88	$0.101 \pm 0.000 \ (\times 10^{-3})$
$1.3 \leq \eta < 1.7$	12	0.000 ± 0.022	5099.56	4589.61	$0.126 \pm 0.000 \ (\times 10^{-3})$
$1.7 \leq \eta < 2.5$	13	0.000 ± 0.022	1910.93	1737.21	$0.080 \pm 0.000 \ (\times 10^{-3})$
$2.5 \leq \eta < 2.8$	11	0.248 ± 0.003	26.01	166.41	$0.100 \pm 0.001 \ (\times 10^{-3})$

In rosso: $\chi^2/\text{ndf} > 3$, fit poco affidabile

In blu: r_2 , la risoluzione nominale ATLAS ad alto p_T